

PROFILE SUMMARY

- Senior Game Developer with 9 years of experience shipping AAA console and PC action-adventure games across combat, traversal, and open-world systems, specializing in gameplay programming, animation integration, and performance optimization on console.
- Solid technical background across engines (Unreal Engine 5, Unity 6), languages (C++, Blueprints), rendering on DirectX 12 and shader authoring in HLSL, target platforms PS5, Xbox Series X|S, and PC, multiplayer on authoritative client-server with rollback, and live ops covering seasonal content drops, A/B tests, and telemetry-driven balance patches, with strong fundamentals in frame-budget discipline, data-oriented design, and a designer-friendly tools mindset.
- Deep expertise in gameplay systems and combat-state machines, ECS / component-based architecture and data-oriented design, shader authoring and post-processing in HLSL, and frame-pacing and memory budget on console, applying practices such as navmesh, behavior trees, and animation state machines and editor-tools authoring for designers to deliver ship-quality, perf-clean, and designer-friendly gameplay code.
- Engaged collaborator working cross-functionally with Design, Art, Animation, Audio, QA, and Production in milestone-paced production with weekly playtests, contributing to gameplay-design reviews, perf war rooms, and post-launch retrospectives with a pragmatic, ship-it-then-polish mindset.
- Senior engineer who shares technical excellence and fosters a culture of code-review rigor and playtest-driven iteration discipline through code-review coaching and tool-building, while running weekly gameplay-engineering guild sessions and authoring widely adopted gameplay-pattern and tools handbooks.

TECHNICAL SKILLS

Engines & Languages:	Unreal Engine 5 (C++ + Blueprints), Unity 6 (C#), Godot 4 (GDScript / C#), proprietary engines, GitHub, Perforce
Gameplay & Systems:	Combat state machines, traversal, weapon systems, ECS, component-based architecture, data-oriented design, gameplay tags
Graphics & Rendering:	HLSL / GLSL shaders, PBR materials, Lumen / Nanite, post-processing, DirectX 12, Vulkan, Metal, ray tracing fundamentals
Physics, Animation & AI:	Chaos / PhysX, animation state machines, IK, navmesh + A* pathfinding, behavior trees, finite state machines, blackboards
Performance & Profiling:	Unreal Insights, Unity Profiler, RenderDoc, PIX, RGP, frame-pacing, LOD authoring, draw-call batching, GPU memory budget
Multiplayer & Networking:	Replication, lag compensation, rollback netcode, dedicated-server hosting, matchmaking, anti-cheat fundamentals
Tools & Pipeline:	Editor extensions, Blueprint nodes, asset import pipelines, Houdini / Maya bridges, Python automation, build automation
Platforms & Live Ops:	PS5 + Xbox Series X S + PC Steam, Switch (cross-vendor study), Steamworks, PSN, Xbox Live, telemetry pipelines, A/B testing

EDUCATION

DigiPen Institute of Technology **B.S. in Computer Science in Real-Time Interactive Simulation**

Redmond, WA • Sep 2013 - Jun 2017

WORK EXPERIENCE

Insomniac Games **Senior Game Developer**

Burbank, CA • Sep 2020 - Present

- Own gameplay programming on **Marvel's Wolverine** for **PS5**, covering the **traversal**, **melee combat**, and **gadget systems** pillars from prototype through cert; partner with **14** gameplay engineers across **3** pods on combat tuning, animation hand-offs, and weekly playtest reviews with design and QA.
- Designed the gameplay systems around **component-based ECS** with **gameplay-tag dispatch**, scaling cleanly to **120+** gameplay components and **40** ability classes; cut designer iteration loops from **30s** to **4s** through hot-reload and data-driven authoring.
- Authored **character shaders** and the **post-processing stack** in **HLSL** with **PBR** materials and screen-space effects, profiling on RenderDoc and PIX; delivered **2.4 ms** of GPU-time savings per frame on **PS5** across the lighting and bloom passes.
- Built enemy **behavior trees**, **navmesh A*** pathing, and **full-body IK** for combat encounters, tuning blackboards and animation state machines against weekly design reviews; shipped **20+** enemy archetypes across **8** mission types in the campaign.
- Drove the perf pass to a locked **60 FPS** on **PS5 performance mode**, pulling frame time from **19.4 ms** to **16.2 ms** through **Unreal Insights** traces, LOD authoring, and draw-call batching; cleared all **14** review-flagged scenes ahead of cert.
- Owned **co-op replication** and the **lag compensation** pass across dedicated servers, with rewind buffers and authority arbitration on the gameplay simulation; lifted hit-reg accuracy from **87%** to **98%** at **120 ms** RTT across the launch window.
- Built editor extensions for the **ability-authoring editor** and the **combat-tuning sliders**, with Blueprint nodes and Python automation behind the import pipeline; adopted by **12** designers across **3** disciplines and cited in **2** postmortems as a top-3 production accelerator.

Riot Games **Game Developer**

Los Angeles, CA • Jul 2017 - Aug 2020

- Partnered with **Design, Art, Animation, Audio, and Production** across the live-service flow, contributing to **3** champions and **2** game modes shipped to live over **9** release cycles in the period.
- Owned **telemetry events** and the **A/B test framework** for the gameplay team, running **18+** live experiments with statistical-rigor reviews; lifted **D7 retention** by **6.2%** on a flagship mode through tuned drop tables and pacing.
- Mentored **3** mid-career engineers through senior gameplay work, running pairing sessions on combat and animation systems; supported **2** promotions to Senior and authored the team's **gameplay code-review handbook** adopted across **4** squads.
- Partnered with **Design, Art, Animation, Audio, QA, and Production** across **14** patches and **4** seasonal events, authoring **22** release-readiness and live-ops runbooks that became the team's release baseline; onboarded **5** new gameplay engineers into the cadence.